

MICHAEL MILLS

+1 (207) 712-7052 | mmills6060@gmail.com | New York, New York | linkedin.com/in/michael-a-mills/ | github.com/mmills6060

EDUCATION

Northeastern University
Master's, Computer Science

January 2023 - June 2025

Gettysburg College
Bachelor's, Organization and Management Studies

August 2016 - March 2020

PROFESSIONAL EXPERIENCE

Athena Intelligence
Forward Deployed Engineer

New York, NY, USA
July 2025 - July 2025

- Delivered full-stack functionality—data ingestion pipelines, AI/ML inference services, and React/Next.js front-ends written in Typescript—to solve client-specific problems and ship production-grade applications at speed.
- Partnered with enterprise clients across legal, finance, defense, supply-chain, retail and media sectors to map high-value workflows and design AI-first solutions with the Olympus analytics platform and Athena digital coworker.
- Built first-of-its-kind product features directly on customer sites (“engineering at the edge”); validated in production, then rolled successful capabilities back into the core code-base for fleet-wide use.
- Integrated Athena’s AI agents with existing enterprise systems and secured environments, driving rapid user adoption of AI-native workflows.

VETRO

Software Engineer

Remote
January 2025 - July 2025

- #1 GitHub contributor to a flagship product that generated \$2.4 million in revenue within the first six months of launch, demonstrating impactful contributions to high-growth software projects and visions.
- Delivered full-stack functionality—data ingestion pipelines, AI/ML inference services, and React/Next.js front-ends written in Typescript—to solve client-specific problems and ship production-grade applications at speed.
- Developed and optimized network services and APIs written in Python/GoLang that interact with core TCP/IP networking layers.
- Worked closely with SREs, DevOps, and Network Engineers to optimize infrastructure performance.
- Managed and maintained Linux-based networking systems, ensuring high availability and performance.

VETRO

Software Engineer Co op

Remote
August 2024 - January 2025

- Leveraged machine learning frameworks such as TensorFlow, PyTorch, and Scikit-Learn, along with Kubeflow for model deployment, to develop and optimize large language models (LLMs), recommendation systems, and custom AI solutions tailored for production environments. These models enhanced product functionality and improved user experience by delivering accurate and scalable AI-driven recommendations and insights.
- Collaborated with a scrum team in an Agile environment to develop, refine, and ship full-stack applications using modern frameworks (React.js, Vue.js, Express.js) and languages (Typescript, Python)
- Gained experience in building and querying geospatial databases using PostgreSQL and PostGIS.
- Developed scalable and fault-tolerant systems by collaborating on designing distributed storage solutions in a large distributed computing environment.

Banbury

Software Development Engineer

Remote
April 2024 - August 2024

- Developed a prototype for a pioneering platform that merges advanced networking techniques with machine learning using Golang, Python, C++, Java, Javascript
- Enhanced platform scalability and functionality by integrating full-stack development principles, developing both front-end and back-end components using SQL databases for data storage and retrieval.
- Improved platform capabilities by implementing BI tools and SQL queries to generate insightful reports for client customization, data setup, configuration, and ongoing maintenance.
- Enhanced scalability and deployment of the platform by utilizing Kubernetes to manage and deploy containerized applications within the decentralized cloud computing platform.
- Facilitated seamless communication between devices by developing a complex networking framework and utilizing Nginx to manage and optimize network traffic, making local AI inferences feasible.

Contract

Software Development Engineer

Remote

August 2023 - April 2024

- Contributed to Agile development and CI/CD practices by collaborating daily with senior software engineers, implementing CI/CD pipelines to streamline development and deployment processes.
- Improved AI inference efficiency by leveraging networking technologies such as DNS and HTTP, ensuring rapid and reliable data access across distributed systems.
- Leveraged relational databases for data management, effectively supporting AI-driven applications
- Created an extensive API library and a Software Development Kit (SDK), facilitating future enhancements and customization, ensuring scalability and adaptability of the technology.

UBS

Registered Client Service Associate

New York, NY, USA

October 2021 - January 2023

- Advanced to a senior role within the Client Service Associate team, specializing in overseeing comprehensive compliance-related issues, including anti-money laundering protocols and financial advisor performance.
- Enhanced business analytics and business intelligence by driving initiatives to improve business processes, supporting compliance and operational efficiency.
- Enhanced audit accuracy and compliance by ensuring adherence to internal operating procedures during audits of accounts and maintenance changes, utilizing SQL for database management.
- Provided strategic advice to regional directors on prevailing trends, highlighting areas for improvement and driving initiatives that enhanced overall business efficiency and compliance standards.
- Increased data-driven decision making by utilizing BI tools and statistics to analyze compliance trends, providing strategic advice and highlighting areas for improvement.

UBS

Financial Advisor

New York, NY, USA

March 2021 - October 2021

- Built relationships with clients to meet business objectives including asset consolidation, diversification, and client deepening
- Improved client understanding of financial markets and economic conditions by utilizing strong communication skills to explain complex concepts.
- Recommended appropriate UBS products and services to meet clients' financial goals by utilizing consultative selling skills
- Analyzed portfolio performance reports, in conjunction with Wealth Management research and external parties, to make suitable investment recommendations

RESEARCH AND PUBLICATIONS

Future-Proofing Computer Behavior: A Framework for Predictive Analytics and Seamless Device Integration

Remote

April 2024 - August 2024

- Utilized predictive analytics to enhance the understanding and management of device behaviors and performance metrics, including Wi-Fi upload/download speeds, GPU usage, CPU usage, and RAM usage.
- Developed and implemented a relay server strategy to enable seamless and secure communication between devices across different network environments.
- Built and trained a Long Short-Term Memory (LSTM) network model to forecast device performance metrics and facilitate informed decision-making for resource allocation.
- Analyzed the performance of predictive models using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) to ensure high accuracy in predictions.

SKILLS

Skills: Node.js, AWS, Computer Vision, C#, Django, Docker, Flask, Heroku, JavaScript, jQuery, Git, HTML/CSS, Java, Keras, Linux/Unix, C/C++, MongoDB, MySQL, .NET, NumPy, Pandas, Postman, Pytorch, React.js, React Native, REST APIs, Ruby, Ruby on Rails, Rust, Salesforce, Python, Go, Excel/Numbers/Sheets, Express.js, Firebase